Run report: which guesses decide degree-3 refutation at m = 4

- **Study:** `research/dreg_m4_guess_bits_20261106/`, pre-registered before any run.
- **Run:** 2026-11-06 22:18 UTC to 22:55 UTC.
- **Class:** stage diagnostic. No `S`, no rho ratio, no speed claim.
- **Standing:** unverified assertion until a pcc verdict exists.

What ran

- **Phase 1:** the 3571 design rerun (arms I and R, ten guesses, all 1024 branches, 8 draws at ℓ = 5), 16,384 runs at degree 3.
- **Analysis:** for each arm and draw, the essential set E: the guesses whose value changes refutation on some branch.
- **Phase 2:** only E guessed, every assignment, 424 runs.

Controls

- The phase-1 records pass the full 3571 scorer, and its per-draw counts reproduce 3571's line for line.
- Every phase-2 record matches its reduced guesses, with `identity_check` true.

Measured values

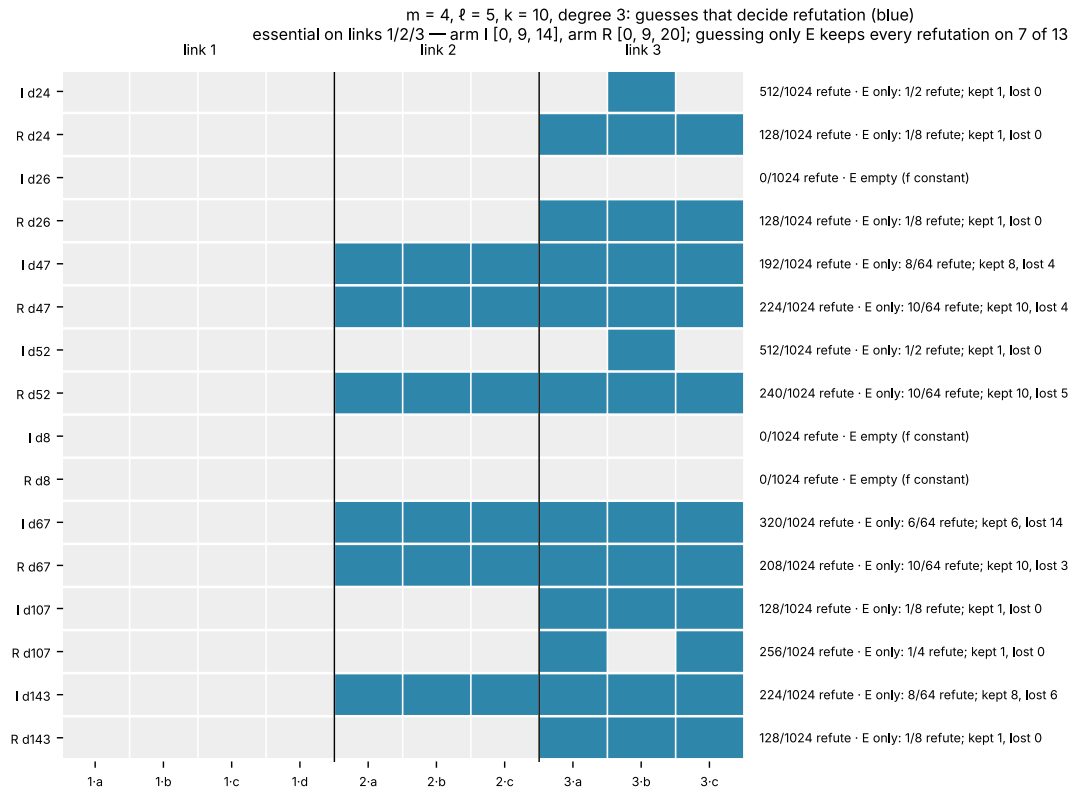


Figure 1: Essential guesses (blue) per arm and draw; right, the refuting count with ten guesses and the E-only phase. Source: `plot.py` (layout only) on `plot-data.json` from `dreg_m4_guess_bits --plot-data score`.

- No link-1 guess is essential on any arm-draw. Essential guesses by link: arm I 0, 9, 14; arm R 0, 9, 20.
- |E| is 0 to 6 everywhere.
- Guessing only E keeps every refutation on 7 of 13 tested arm-draws, all those with $|E| \le 3$, and gains none anywhere.